

Algebra 1
Unit 1
Lesson 4 Practice

Name **Answer Key** _____
Date _____
Period _____

For questions 1-5, use the exponent laws to write equivalent expressions.

1. $\left(\frac{6}{5}\right)^{-2} = \frac{6^{-2}}{5^{-2}} \rightarrow \frac{5^2}{6^2}$

2. $\frac{7^7}{7^4} = 7^{7-4} = 7^3$

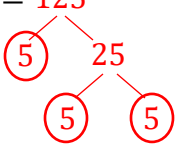
3. $(5.2 \cdot (-9))^3 = 5.2^3 \cdot (-9)^3$

4. $(-2) \cdot (-2)^{-5} \cdot (-2)^8 = (-2)^{1-5+8} = (-2)^4$

5. $((-4)^4)^9 = (-4)^{4 \cdot 9} = (-4)^{36}$

Find the value of each for questions 6-7.

6. $\sqrt[3]{125} = 125$ $\sqrt[3]{5 \cdot 5 \cdot 5} = 5$



7. $\sqrt{25} = 25$ $\sqrt{5 \cdot 5} = 5$



8. Explain why both the $\sqrt{25}$ and $\sqrt[3]{125}$ both equal 5.
Because the square root of 25 is 5, $5 \cdot 5 = 25$ and the cubed root of 125 is 5, $5 \cdot 5 \cdot 5 = 125$. Two of the same factor is the square root and three of the same factor is the cubed root.

For questions 9-10, find the value of the variable that makes the equation true.

9. $\sqrt{c} \cdot \sqrt{c} = 11$ $\sqrt{11} \cdot \sqrt{11} = \sqrt{121} = 11$

$c = \underline{\underline{11}}$

10. $\frac{2^3}{3^3} = \left(\frac{3}{2}\right)^g$ $\frac{3^{-3}}{2^{-3}} = \frac{2^3}{3^3}$

$g = \underline{\underline{-3}}$